Lectures on

Complex variables

Delivered By

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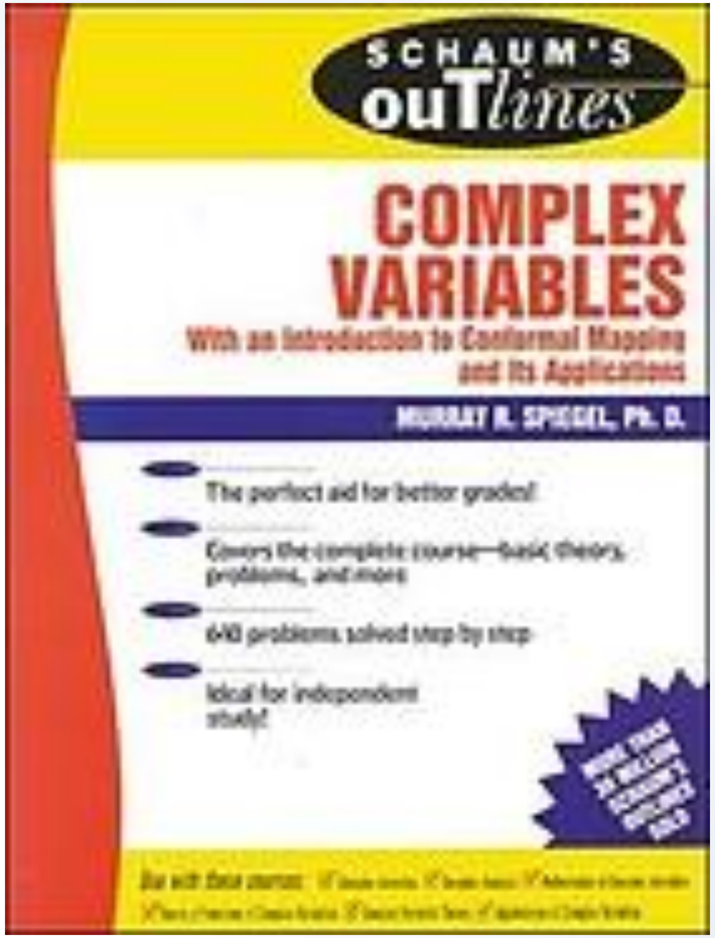


6.1 Sequences of Functions

The ideas of Chapter 2, pages 48 and 49, for sequences and series of constants are easily extended to sequences and series of functions.

Let $u_1(z), u_2(z), \ldots, u_n(z), \ldots,$ denoted briefly by $\{u_n(z)\}$, be a sequence of functions of z defined and single-valued in some region of the z plane. We call $U(z)$ the *limit* of $u_n(z)$ as $n \to \infty$, and write $\lim_{n\to\infty} u_n(z) = U(z)$, if given any positive number ϵ, we can find a number N [depending in general on both ϵ and z] such that

$$|u_n(z) - U(z)| < \epsilon \quad \text{for all } n > N$$

In such a case, we say that the sequence *converges* or is *convergent* to $U(z)$.

If a sequence converges for all values of z (points) in a region $\mathcal{R}$, we call $\mathcal{R}$ the *region of convergence* of the sequence. A sequence that is not convergent at some value (point) z is called *divergent* at z.

The theorems on limits given on page 49 can be extended to sequences of functions.

6.2 Series of Functions

From the sequence of functions $\{u_n(z)\}$, let us form a new sequence $\{S_n(z)\}$ defined by

$$\begin{aligned} S_1(z) &= u_1(z) \\ S_2(z) &= u_1(z) + u_2(z) \\ &\vdots \\ S_n(z) &= u_1(z) + u_2(z) + \cdots + u_n(z) \end{aligned}$$

where $S_n(z)$, called the nth *partial sum*, is the sum of the first n terms of the sequence $\{u_n(z)\}$.

The sequence $S_1(z), S_2(z), \ldots$ or $\{S_n(z)\}$ is symbolized by

$$u_1(z) + u_2(z) + \cdots = \sum_{n=1}^{\infty} u_n(z) \tag{6.1}$$

called an *infinite series*. If $\lim_{n\to\infty} S_n(z) = S(z)$, the series is called *convergent* and $S(z)$ is its *sum*; otherwise, the series is called *divergent*. We sometimes write $\sum_{n=1}^{\infty} u_n(z)$ as $\sum u_n(z)$ or $\sum u_n$ for brevity.

6.3 Absolute Convergence

A series $\sum_{n=1}^{\infty} u_n(z)$ is called *absolutely convergent* if the series of absolute values, i.e., $\sum_{n=1}^{\infty} |u_n(z)|$, converges.

If $\sum_{n=1}^{\infty} u_n(z)$ converges but $\sum_{n=1}^{\infty} |u_n(z)|$ does not converge, we call $\sum_{n=1}^{\infty} u_n(z)$ *conditionally convergent*.

6.5 Power Series

A series having the form

$$a_0 + a_1(z-a) + a_2(z-a)^2 + \cdots = \sum_{n=0}^{\infty} a_n(z-a)^n \tag{6.2}$$

is called a *power series* in $z - a$. We shall sometimes shorten (6.2) to $\sum a_n(z-a)^n$.

Clearly the power series (6.2) converges for $z = a$, and this may indeed be the only point for which it converges [see Problem 6.13(b)]. In general, however, the series converges for other points as well. In such cases, we can show that there exists a positive number R such that (6.2) converges for $|z-a| < R$ and diverges for $|z-a| > R$, while for $|z-a| = R$, it may or may not converge.

Geometrically, if Γ is a circle of radius R with center at $z = a$, then the series (6.2) converges at all points inside Γ and diverges at all points outside Γ, while it may or may not converge on the circle Γ. We can consider the special cases $R = 0$ and $R = \infty$, respectively, to be the cases where (6.2) converges only at $z = a$ or converges for all (finite) values of z. Because of this geometrical interpretation, R is often called the *radius of convergence* of (6.2) and the corresponding circle is called the *circle of convergence*.

B. Theorems on Absolute Convergence

THEOREM 6.8. If $\sum_{n=1}^{\infty} |u_n|$ converges, then $\sum_{n=1}^{\infty} u_n$ converges. In words, an absolutely convergent series is convergent.

THEOREM 6.9. The terms of an absolutely convergent series can be rearranged in any order and all such rearranged series converge to the same sum. Also, the sum, difference, and product of absolutely convergent series is absolutely convergent.

These are not so for conditionally convergent series (see Problem 6.127).

C. Special Tests for Convergence

THEOREM 6.10. (*Comparison tests*)

(a) If $\sum |v_n|$ converges and $|u_n| \le |v_n|$, then $\sum u_n$ converges absolutely.

(b) If $\sum |v_n|$ diverges and $|u_n| \ge |v_n|$, then $\sum |u_n|$ diverges but $\sum u_n$ may or may not converge.

THEOREM 6.11. (*Ratio test*) Let $\lim_{n\to\infty} |u_{n+1}/u_n| = L$. Then $\sum u_n$ converges (absolutely) if $L < 1$ and diverges if $L > 1$. If $L = 1$, the test fails.

THEOREM 6.12. (*n*th *Root test*) Let $\lim_{n\to\infty} \sqrt[n]{|u_n|} = L$. Then $\sum u_n$ converges (absolutely) if $L < 1$ and diverges if $L > 1$. If $L = 1$, the test fails.

THEOREM 6.13. (*Integral test*) If $f(x) \ge 0$ for $x \ge a$, then $\sum f(n)$ converges or diverges according as $\lim_{M\to\infty} \int_a^M f(x)\,dx$ converges or diverges.

THEOREM 6.14. (*Raabe's test*) Let $\lim_{n\to\infty} n\left(1 - |u_{n+1}/u_n|\right) = L$. Then $\sum u_n$ converges (absolutely) if $L > 1$ and diverges or converges conditionally if $L < 1$. If $L = 1$, the test fails.

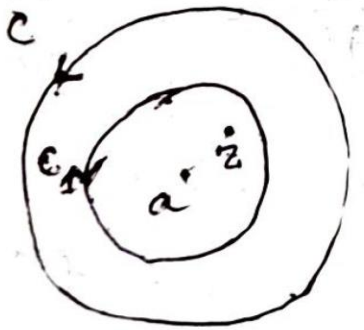
THEOREM 6.15. (*Gauss' test*) Suppose $|u_{n+1}/u_n| = 1 - (L/n) + (c_n/n^2)$ where $|c_n| < M$ for all $n > N$. Then $\sum u_n$ converges (absolutely) if $L > 1$ and diverges or converges conditionally if $L \le 1$.

THEOREM 6.16. (*Alternating series test*) If $a_n \ge 0$, $a_{n+1} \le a_n$ for $n = 1, 2, 3, \ldots$ and $\lim_{n\to\infty} a_n = 0$, then $a_1 - a_2 + a_3 - \cdots = \sum (-1)^{n-1} a_n$ converges.

Taylor's Theorem

Statement: If $f(z)$ is analytic inside a circle C with centre at a then for all z inside C,

$$f(z) = f(a) + f'(a)(z-a) + \frac{f''(a)}{2!}(z-a)^2 + \frac{f'''(a)}{3!}(z-a)^3 + \cdots$$



State and proof laurents theorem.

Statement : Suppose $f(z)$ is analytic inside and on the boundary of the ring shaped region R bounded by two concentric circles C_1 and C_2 with centre at a and respective radii r_1 and r_2 $(r_1 > r_2)$.

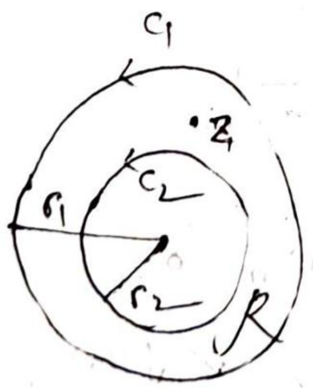
Then for all z in R.

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{a_{-n}}{(z-a)^n}$$

where,

$$a_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(\omega)}{(\omega-a)^{n+1}}\, d\omega \quad , n = 0, 1, 2, \ldots$$

$$a_{-n} = \frac{1}{2\pi i} \oint_{C_2} \frac{f(\omega)}{(\omega-a)^{-n+1}}\, d\omega \quad , n = 1, 2, \ldots$$



Let $f(z) = \ln(1+z)$, where we consider that branch which has the value zero when $z = 0$.

(a) Expand $f(z)$ in a Taylor series about $z = 0$

Expand $\ln\left(\frac{1+z}{1-z}\right)$ in a Taylor series about $z = 0$.

Soln: (a) Given that,

$$f(z) = \ln(1+z) \qquad \therefore f(0) = 0$$

$$f'(z) = \frac{1}{1+z}, \qquad \therefore f'(0) = 1$$

$$f''(z) = -\frac{1}{(1+z)^2}, \qquad \therefore f''(0) = -1$$

$$f'''(z) = (-1)(-2)\frac{1}{(1+z)^3}, \qquad f'''(0) = 2!$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$f^{n+1}(z) = (-1)^n n! \frac{1}{(1+z)^{n+1}}, \qquad f^{n+1}(0) = (-1)^n n!$$

then $f(z) = \ln(1+z) = f(0) + f'(0)z + \frac{f''(0)}{2!}z^2 + \frac{f'''(0)}{3!}z^3 + \cdots$

$$= z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \cdots$$

(c) From the result in (a) we have ; on replacing z by $-z$.

$$\ln(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \cdots$$

$$\ln(1-z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \frac{z^4}{4} + \cdots$$

both series converges for $|z|<1$.

By subtraction, we have.

$$\ln\left(\frac{1+z}{1-z}\right) = 2\left(z + \frac{z^3}{3} + \frac{z^5}{5} + \cdots\right)$$

$$= \sum_{n=0}^{\infty} \frac{2z^{2n+1}}{2n+1}$$

which converges for $|z|<1$, we can also show that this series convergent for $|z|=1$ except for $z=\pm 1$.

Expand $f(z) = \cos z$ in a Taylor's Series about $\pi/2$ respectively.

Soln: We have,

$$f(z) = \cos z \quad , \quad f(\pi/2) = 0$$

$$f'(z) = -\sin z \quad , \quad f'(\pi/2) = -1$$

$$f''(z) = -\cos z \quad , \quad f''(\pi/2) = 0$$

$$f'''(z) = \sin z \quad , \quad f'''(\pi/2) = 1$$

$$f^{iv}(z) = \cos z \quad , \quad f^{iv}(\pi/2) = 0$$

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We know the Taylor series

$$f(z) = f(a) + \frac{f'(a)}{1!}(z-a) + \frac{f''(a)}{2!}(z-a)^2 + \frac{f'''(a)}{3!}(z-a)^3$$

$$+ \cdots\cdots$$

$$f(z) = 0 - (z - \pi/2) + \frac{(z-\pi/2)^3}{3!} - \frac{(z-\pi/2)^5}{5!} + \cdots$$

$$\therefore\ f(z) = -\left(z - \frac{\pi}{2}\right) + \frac{(z-\pi/2)^3}{3!} - \frac{(z-\pi/2)^5}{5!} + \cdots\cdots$$

Ans

Expand $f(z) = \sin z$ in a Taylor series about $z = \pi/4$ and determine the region of converges of this series.

Q.17. Find the Laurent expansion of $f(z) = \frac{1}{1+z}$ for $|z| > 1$.

Soln

given $f(z) = \frac{1}{1+z}$, $|z| > 1$

$$\therefore f(z) = \frac{1}{z\left(1+\frac{1}{z}\right)}$$

$$= \frac{1}{z}\left(1+\frac{1}{z}\right)^{-1}$$

$$= \frac{1}{z}\left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \cdots\right)$$

$$= \frac{1}{z} - \frac{1}{z^2} + \frac{1}{z^3} - \frac{1}{z^4} + \cdots$$

Thus the Laurent series is

$$\frac{1}{1+z} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{z^n} \quad \text{for } |z| > 1.$$

Find the Laurent expansion of

$$f(z) = \frac{1}{(z-1)(z-2)} \text{ for } 1<|z|<2.$$

Solⁿ $f(z) = \frac{1}{(z-1)(z-2)} = \left(\frac{-1}{z-1} + \frac{1}{z-2}\right)$

Now for $\frac{1}{z-1}$ and $|z|>1$

we get

$$\frac{1}{z-1} = \frac{1}{z(1-\frac{1}{z})} = \frac{1}{z}\left(1-\frac{1}{z}\right)^{-1}$$

$$= \frac{1}{z}\left(1 + \frac{1}{z} + \frac{1}{z^2} + \cdots\right)$$

$$= \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots$$

And for $\frac{1}{z-2}$ and $|z| < 2$

we get

$$\frac{1}{z-2} = \frac{1}{-2\left(1-\frac{z}{2}\right)} = \frac{1}{-2}\left(1-\frac{z}{2}\right)^{-1}$$

$$= -\frac{1}{2}\left(1 \overset{-}{+} \frac{z}{2} \overset{+}{+} \frac{z^2}{4} \overset{-}{+} \cdots\cdots\right)$$

Thus the Laurent series of $f(z)$, valid for $1<|z|<2$ is

$$f(z) = -\left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\cdots\right) - \frac{1}{2}\left(1 + \frac{z}{2} + \frac{z^2}{4} + \cdots\right)$$

Expand, $f(z) = \frac{1}{z-3}$ in a Laurent series for the region (i) $|z| < 3$ (ii) $|z| > 3$.

Expan $f(z) = \frac{1}{(z-1)(z-3)}$ in a Laurent series for

(a) $0 < z < 1$

(b) $1 < |z| < 3$

(c) $|z| > 3$.